Level-Set Deficit Identity for the Nicola–Tilli Route

This note implements the first concrete step from `plan2.md`: turn the convexity gap in `faber-krahn-1.pdf` into an exact weighted integral of the two losses in the level-set proof.

Throughout, let $u = u_\Omega$ solve

$$-\Delta u = 1 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega,$$

and set

$$E_t = \{u > t\}, \qquad m(t) = |E_t|, \qquad \Sigma_t = \{u = t\}, \qquad P(t) = \mathcal{H}^{n-1}(\Sigma_t).$$

The formulas below are written for smooth domains and regular levels, to expose the algebra transparently. The rigorous finite-perimeter/coarea version—valid for every open Ω of finite measure, with no smoothness of $\partial\Omega$ and no pointwise lower bound on $|\nabla u|$, all integrals taken on reduced boundaries and holding for a.e. level—is proved in the companion file `WIP_LevelSetIdentity.tex` (Theorem: *Exact deficit identity on reduced-boundary level sets*), via the BV coarea formula, a Lipschitz-truncation flux identity $\int_{\partial^*\{u>t\}} |\nabla u| = m(t)$, the De Giorgi isoperimetric inequality, and the rearranged primitive. That finite-perimeter theorem is the version imported downstream.

1 The Two Pointwise Losses

For a regular level t, define

$$a(t) := -m'(t) = \int_{\Sigma_t} \frac{1}{|\nabla u|}\, d\mathcal{H}^{n-1}.$$

By the divergence theorem applied to E_t,

$$\int_{\Sigma_t} |\nabla u|\, d\mathcal{H}^{n-1} = m(t).$$

Let

$$c_n := n^2 \omega_n^{2/n}.$$

The total level-set loss in the differential inequality is

$$D_{\mathrm{lev}}(t) := a(t)m(t) - c_n m(t)^{2-2/n}.$$

It splits as

$$D_{\mathrm{lev}}(t) = D_H(t) + D_I(t),$$

where

$$D_H(t) := \left(\int_{\Sigma_t} \frac{1}{|\nabla u|}\right)\left(\int_{\Sigma_t} |\nabla u|\right) - P(t)^2 = a(t)m(t) - P(t)^2$$

is the Holder/Cauchy–Schwarz loss, and

$$D_I(t) := P(t)^2 - c_n m(t)^{2-2/n}$$

is the squared-perimeter isoperimetric loss of E_t.

Both terms are nonnegative: $D_H(t) \geq 0$ by Cauchy–Schwarz, and $D_I(t) \geq 0$ by the isoperimetric inequality.

2 Convexity Variable and Exact Identity

Let $L := |\Omega|$, let $v(s) = u^*(s) = m^{-1}(s)$, and define

$$I(s) := \int_{\{u>v(s)\}} u\,dx.$$

Equivalently, for regular values,

$$I'(s) = v(s), \qquad I''(s) = v'(s) = \frac{1}{m'(v(s))} = -\frac{1}{a(v(s))}.$$

The function used in `faber-krahn-1.pdf` is

$$G(s) := I(s) + \frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}}.$$

Since

$$\frac{d^2}{ds^2}\frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}} = \frac{1}{c_n s^{1-2/n}},$$

we get, with $s = m(t)$,

$$G''(s) = \frac{1}{c_n s^{1-2/n}} - \frac{1}{a(t)} = \frac{D_{\mathrm{lev}}(t)}{s\,a(t)\,c_n s^{1-2/n}}.$$

The endpoint convexity gap is

$$\Gamma(\Omega) := G'(L) - \frac{G(L)}{L}.$$

Because $G(0) = 0$,

$$\Gamma(\Omega) = \frac{1}{L}\int_0^L sG''(s)\,ds.$$

Changing variables $s = m(t)$, so that $ds = -a(t)\,dt$, gives the exact deficit identity

$$\boxed{\Gamma(\Omega) = \frac{1}{|\Omega|}\int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2\omega_n^{2/n} m(t)^{1-2/n}}\,dt.}$$

This is the clean quantitative form of the rigidity statement at the end of `faber-krahn-1.pdf`.

3 Relation to the Saint-Venant Deficit

At $s = L$, one has $v(L) = 0$. Hence

$$G'(L) = \frac{L^{2/n}}{2n\omega_n^{2/n}}.$$

Also

$$\frac{G(L)}{L} = \frac{1}{L}\int_\Omega u\,dx + \frac{L^{2/n}}{(4+2n)\omega_n^{2/n}}.$$

Therefore

$$\Gamma(\Omega) = \frac{1}{L}\left(\int_B u_B\,dx - \int_\Omega u_\Omega\,dx\right),$$

where B is the ball with $|B| = |\Omega|$. Since

$$E(\Omega) = -\frac{1}{2} \int_\Omega u_\Omega \, dx,$$

we obtain

$$\boxed{\Gamma(\Omega) = \frac{2}{|\Omega|} \big(E(\Omega) - E(B)\big).}$$

Thus the endpoint convexity gap is not merely comparable to the torsion deficit; it is exactly the normalized Saint-Venant deficit.

4 Profile-Gap Monotonicity Inspired by the STFT Stability Paper

The paper `s00222-024-01248-2-3.pdf` proves stability of the Nicola–Tilli STFT theorem by studying the area between the rearranged profile $u^*(s)$ and the optimizer profile e^{-s}. There is a direct torsion analogue.

Let B be the ball with $|B| = |\Omega| =: L$, and let

$$b(s) := u_B^*(s), \qquad v(s) := u_\Omega^*(s).$$

For a ball of radius R, $L = \omega_n R^n$, one has

$$b(s) = \frac{R^2 - (s/\omega_n)^{2/n}}{2n}, \qquad 0 \le s \le L,$$

and therefore

$$b'(s) = -\frac{1}{n^2 \omega_n^{2/n} s^{1-2/n}}.$$

The level-set differential inequality gives

$$v'(s) \ge b'(s)$$

for a.e. $s \in (0, L)$. Hence

$$h(s) := b(s) - v(s)$$

is nonnegative and nonincreasing, with $h(L) = 0$. Moreover

$$\int_0^L h(s)\, ds = \int_B u_B \, dx - \int_\Omega u_\Omega \, dx = 2(E(\Omega) - E(B)).$$

Consequently, for every $s \in (0, L]$,

$$0 \le b(s) - v(s) \le \frac{2(E(\Omega) - E(B))}{s}.$$

In particular, on every boundary-volume slab $s \in [\theta L, L]$,

$$\|b - v\|_{L^\infty([\theta L, L])} \le \frac{2(E(\Omega) - E(B))}{\theta L}.$$

This is a useful quantitative replacement for compactness at the one-dimensional profile level. It says that small Saint-Venant deficit forces the torsion level corresponding to a prescribed near-boundary volume $s \sim L$ to be close to the ball's corresponding level.

For example, if $s = L - \eta$ and $\eta \ll L$, then

$$b(L-\eta) = \frac{R^2 - ((L-\eta)/\omega_n)^{2/n}}{2n} \simeq_{n,L} \eta.$$

Thus if $E(\Omega) - E(B) \ll \eta$, the level

$$t_\eta := v(L-\eta)$$

is positive and comparable to the ball boundary-layer height. This provides a candidate near-boundary level E_{t_η} with exactly

$$|\Omega \setminus E_{t_\eta}| = \eta.$$

The missing part is still to make sure this same level has small geometric level-set deficit. The exact identity above gives averaged control, and the hybrid strategy is to combine this profile control with selected-minimizer regularity to extract good near-boundary levels.

5 Consequences for Good Levels

The quantitative isoperimetric inequality gives, for a dimensional constant $c > 0$,

$$D_I(t) \geq c\, m(t)^{2-2/n} \mathcal{A}(E_t)^2$$

for a.e. regular level t. Hence

$$E(\Omega) - E(B) \gtrsim_n \int_0^{\|u\|_\infty} m(t)\mathcal{A}(E_t)^2\, dt$$

up to the normalization factor $|\Omega|$. More precisely, the exact identity controls the weighted average

$$\int_0^{\|u\|_\infty} \frac{D_I(t)}{m(t)^{1-2/n}}\, dt.$$

For any measurable level interval $J \subset (0, \|u\|_\infty)$, define the weight

$$d\nu(t) := \frac{dt}{n^2\omega_n^{2/n} m(t)^{1-2/n}}.$$

If $\nu(J) > 0$, then there exists a regular $t \in J$ such that

$$D_H(t) + D_I(t) \leq \frac{|\Omega|\Gamma(\Omega)}{\nu(J)}.$$

In particular, on any slab where $m(t)$ is bounded above and below, the Saint-Venant deficit produces at least one level with both small isoperimetric loss and small Holder loss.

6 Holder Loss as a Serrin-Variance Defect

Let $f = |\nabla u|$ on Σ_t, and set

$$\bar{f} := \frac{1}{P(t)} \int_{\Sigma_t} f = \frac{m(t)}{P(t)}.$$

Then the Holder deficit has the exact weighted variance representation

$$\boxed{\int_{\Sigma_t} \frac{(f-\bar{f})^2}{f}\, d\mathcal{H}^{n-1} = \frac{m(t)}{P(t)^2} D_H(t).}$$

Indeed,

$$\int_{\Sigma_t} \frac{(f-\bar{f})^2}{f} = \int_{\Sigma_t} f - 2\bar{f}P(t) + \bar{f}^2 \int_{\Sigma_t} \frac{1}{f} = \frac{m(t)}{P(t)^2}\left(m(t) \int_{\Sigma_t} \frac{1}{f} - P(t)^2 \right).$$

Thus small $D_H(t)$ says that $|\nabla u|$ is almost constant on Σ_t, in the natural weighted $L^2(1/f)$ sense.

If, on a good regular level, one also has

$$0 < c_t \leq |\nabla u| \leq C_t < \infty,$$

then this becomes an ordinary L^2-variance estimate:

$$\int_{\Sigma_t} |f-\bar{f}|^2 \, d\mathcal{H}^{n-1} \leq C_t \frac{m(t)}{P(t)^2} D_H(t).$$

This is the precise bridge from the Nicola–Tilli convexity gap to an approximate Serrin condition on the interior domain E_t.

7 What Remains Open in This Route

The identity proves that small Saint-Venant deficit forces many superlevel sets to be simultaneously almost isoperimetric and almost overdetermined.

The unsolved propagation problem is to recover $\mathcal{A}(\Omega)^2$ from this information. The deficit identity sees interior levels with positive height, while Fraenkel asymmetry is a $t=0$ boundary datum. A sharp proof by this route needs a boundary-layer theorem of the following form:

$$\exists t \downarrow 0 \quad \text{such that} \quad |\Omega \setminus E_t| \text{ is small and } D_H(t) + D_I(t) \text{ remains controlled.}$$

Without such a theorem, the level-set method gives strong information on interior cores E_t, but not yet a sharp estimate for the original domain Ω.